

Transactions

ACID Properties

- **Atomicity** - All statements succeed or none succeed.
- **Consistency** - Data moves from one valid state to another.
- **Isolation** - Parallel transactions don't interfere.
- **Durability** - Committed data is permanently saved.

Transactions

Disable autocommit

SET autocommit = 0;

Enable autocommit

SET autocommit = 1;

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Transactions

Start & Commit

START TRANSACTION;

UPDATE *accounts* **SET** *balance* = *balance* - 50 **WHERE** *id* = 1;

UPDATE *accounts* **SET** *balance* = *balance* + 50 **WHERE** *id* = 2;

COMMIT;

Transactions

Rollback

START TRANSACTION;

UPDATE *accounts* **SET** *balance = balance - 100* **WHERE** *id = 1;*

UPDATE *accounts* **SET** *balance = balance + 100* **WHERE** *id = 3;*

ROLLBACK;

Transactions

Savepoint

START TRANSACTION;

UPDATE *accounts* **SET** *balance = balance + 1000* **WHERE** *id = 1;*

SAVEPOINT *after_wallet_topup;*

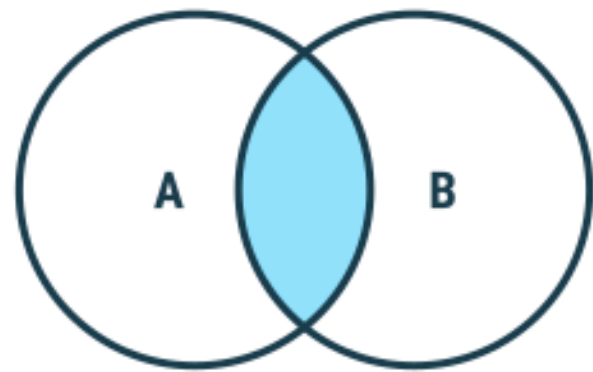
UPDATE *accounts* **SET** *balance = balance + 10* **WHERE** *id = 1;*

ROLLBACK TO *after_wallet_topup;*

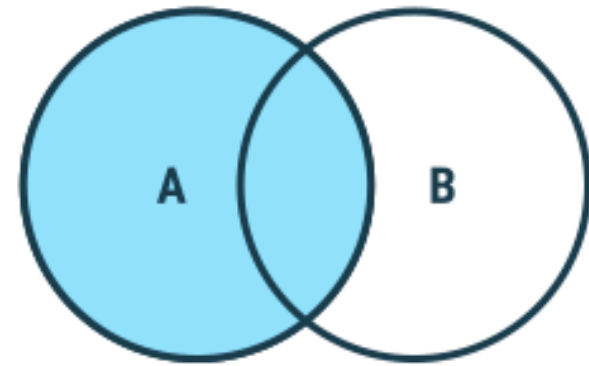
COMMIT;

JOINS

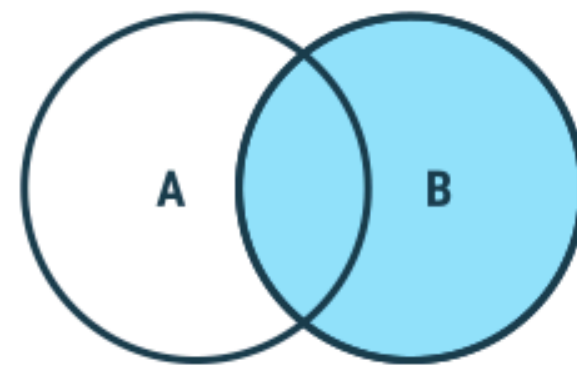
JOINS are used to combine rows from two or more tables based on a related column between them.



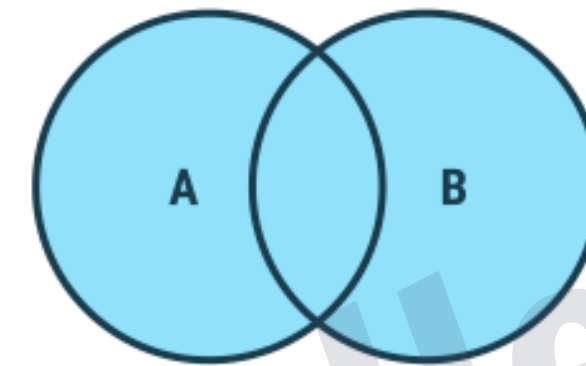
Inner Join



Left Join



Right Join

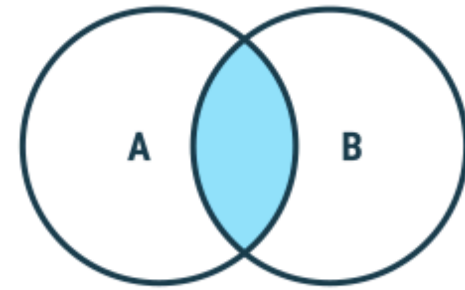


Full Join

Outer Joins

JOINS

INNER JOIN



customer_id	name	city
1	Alice	Mumbai
2	Bob	Delhi
3	Charlie	Bangalore
4	David	Mumbai

customers

order_id	customer_id	amount
101	1	500
102	1	900
103	2	300
104	5	700

orders

Syntax

SELECT *column(s)*

FROM *tableA*

INNER JOIN *tableB*

ON *tableA.col_name = tableB.col_name;*

```
-- inner join
```

```
SELECT c.name, o.order_id, o.amount
```

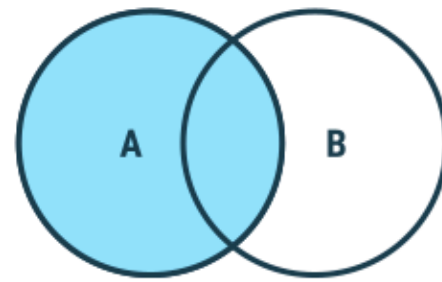
```
FROM customers c
```

```
INNER JOIN orders o
```

```
ON c.customer_id = o.customer_id;
```

JOINS

LEFT JOIN



customer_id	name	city
1	Alice	Mumbai
2	Bob	Delhi
3	Charlie	Bangalore
4	David	Mumbai

customers

order_id	customer_id	amount
101	1	500
102	1	900
103	2	300
104	5	700

orders

Syntax

SELECT *column(s)*

FROM *tableA*

LEFT JOIN *tableB*

ON *tableA.col_name = tableB.col_name;*

```
-- left join
```

```
SELECT *
```

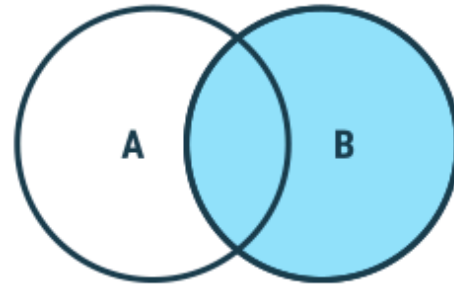
```
FROM customers c
```

```
LEFT JOIN orders o
```

```
ON c.customer_id = o.customer_id;
```

JOINS

RIGHT JOIN



customer_id	name	city
1	Alice	Mumbai
2	Bob	Delhi
3	Charlie	Bangalore
4	David	Mumbai

customers

order_id	customer_id	amount
101	1	500
102	1	900
103	2	300
104	5	700

orders

Syntax

SELECT *column(s)*

FROM *tableA*

RIGHT JOIN *tableB*

ON *tableA.col_name = tableB.col_name;*

```
-- right join
```

```
SELECT *
```

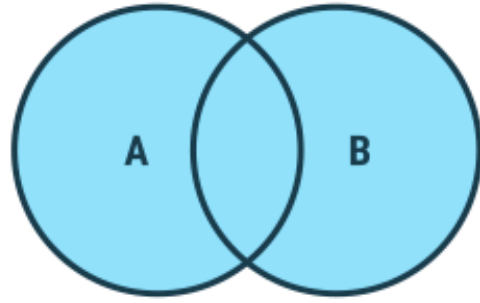
```
FROM customers c
```

```
RIGHT JOIN orders o
```

```
ON c.customer_id = o.customer_id;
```

JOINS

OUTER JOIN



LEFT JOIN

UNION

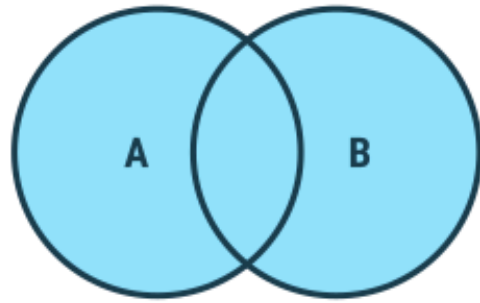
RIGHT JOIN

Syntax in MySQL

```
SELECT * FROM customers as c
LEFT JOIN orders as o
ON c.customer_id = o.customer_id
UNION
SELECT * FROM customers as c
RIGHT JOIN orders as o
ON c.customer_id = o.customer_id;
```

JOINS

CROSS JOIN



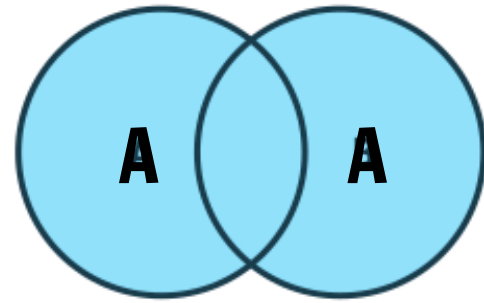
Syntax

```
SELECT column(s)  
FROM tableA  
CROSS JOIN tableB ;
```

```
-- cross join  
SELECT *  
FROM customers as c  
CROSS JOIN orders as o;  
  
-- inner join  
SELECT *  
FROM customers as A  
JOIN customers as B  
ON A.customer_id = B.customer_id;
```

JOINS

SELF JOIN



It is a regular join but the table is joined with itself.

Syntax

SELECT *column(s)*

FROM *table as a*

JOIN *table as b*

ON *a.col_name = b.col_name;*

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Practice Qs

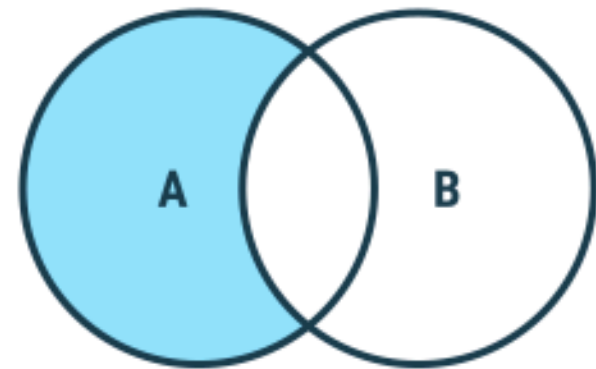
Write SQL command to display the exclusive joins :

customer_id	name	city
1	Alice	Mumbai
2	Bob	Delhi
3	Charlie	Bangalore
4	David	Mumbai

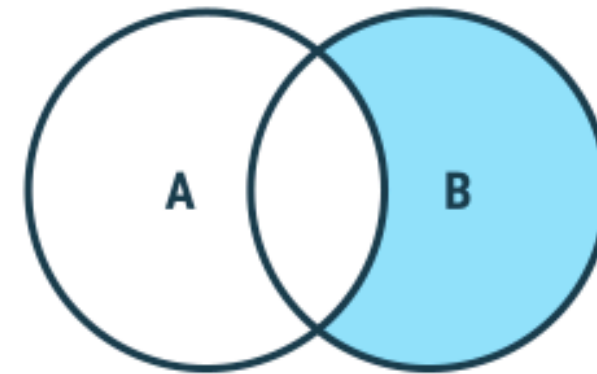
customers

order_id	customer_id	amount
101	1	500
102	1	900
103	2	300
104	5	700

orders



Left Exclusive Join



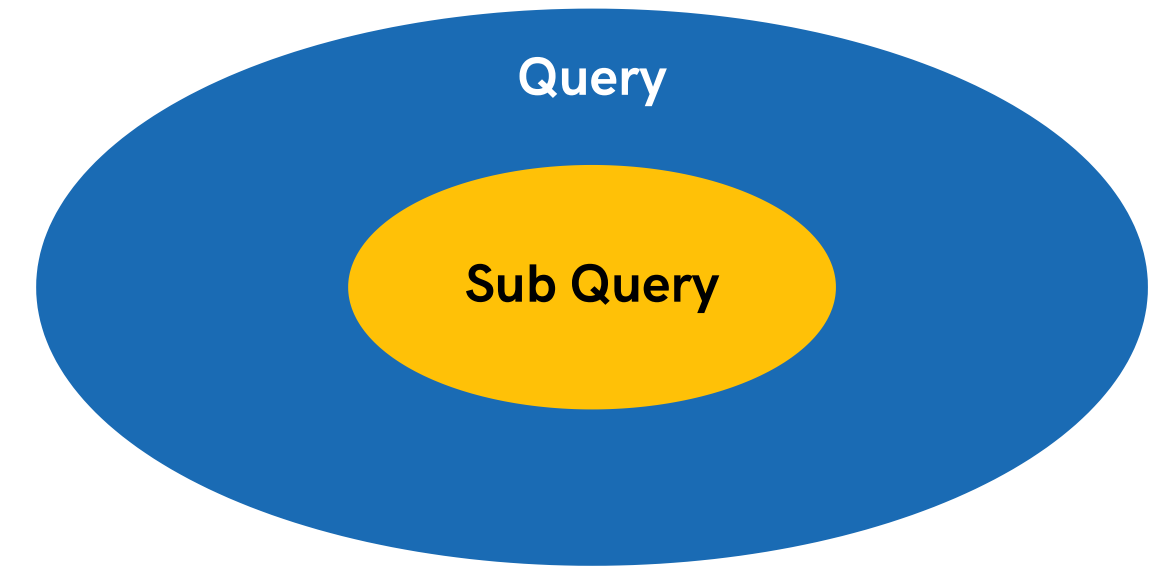
Right Exclusive Join

```
-- left exclusive
SELECT *
FROM customers as c
LEFT JOIN orders as o
ON c.customer_id = o.customer_id
WHERE o.customer_id IS NULL;
```

```
-- right exclusive
SELECT *
FROM customers as c
RIGHT JOIN orders as o
ON c.customer_id = o.customer_id
WHERE c.customer_id IS NULL;
```

Sub-Queries

A Subquery or Inner query or a Nested query is a query within another SQL query. It involves 2 select statements.



Syntax

SELECT *column(s)*

FROM *table_name*

WHERE *col_name operator*

(*subquery*);

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Sub-Queries

With WHERE

```
SELECT *  
FROM orders  
WHERE amount > (  
    SELECT AVG(amount)  
    FROM orders  
);
```

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Sub-Queries

With SELECT

```
SELECT name,  
       ( SELECT COUNT(*)  
         FROM orders o  
         WHERE o.customer_id = c.customer_id  
       ) as order_count  
FROM customers c;
```

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Sub-Queries

With FROM

```
SELECT
    summary.customer_id,
    summary.avg_amount
FROM
    (
        SELECT
            customer_id,
            AVG(amount) AS avg_amount
        FROM orders
        GROUP BY customer_id
    ) AS summary;
```

Views in SQL

A view is a virtual table based on the result-set of an SQL statement.

Syntax

```
CREATE VIEW view1 AS  
SELECT col1, col2 FROM table_name;
```

*A view always shows up-to-date data.
The database engine recreates the view,
every time a user queries it.

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Views in SQL

- No data is stored physically (unless it's a materialized view in some DBs).
- Can include columns from one or more tables.
- Can be used in SELECT, JOIN, or even WHERE clauses like a normal table.
- Helps with security by exposing only certain columns to users.

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Index in SQL

indexes are special database objects that make **data retrieval faster**.

Syntax (single col & multi-col)

CREATE INDEX *idx_name* ON *table*(*col*);

CREATE INDEX *idx_name* ON *table*(*col1*, *col2*);

SHOW INDEX FROM *table*;

DROP INDEX *idx_name* ON *table*;

Stored Procedures

Predefined set of SQL statements that you can save in the database and execute whenever needed.

Syntax (Create)

CREATE PROCEDURE *procedure_name* (*parameters*)

BEGIN

-- SQL statements

END;

```
DELIMITER $$
```

```
CREATE PROCEDURE check_balance(IN acc_id INT, OUT bal DECIMAL(10, 2))
```

```
BEGIN
```

```
    SELECT balance INTO bal
```

```
    FROM bank_accounts as b
```

```
    WHERE b.account_id = acc_id;
```

```
END $$
```

```
DELIMITER ;
```

```
CALL check_balance(2, @balance);
```

```
SELECT @balance;
```

Stored Procedures

Syntax (Call)

CALL *procedure_name* (*arguments*);

Syntax (Drop)

DROP PROCEDURE **IF EXISTS** *procedure_name*;